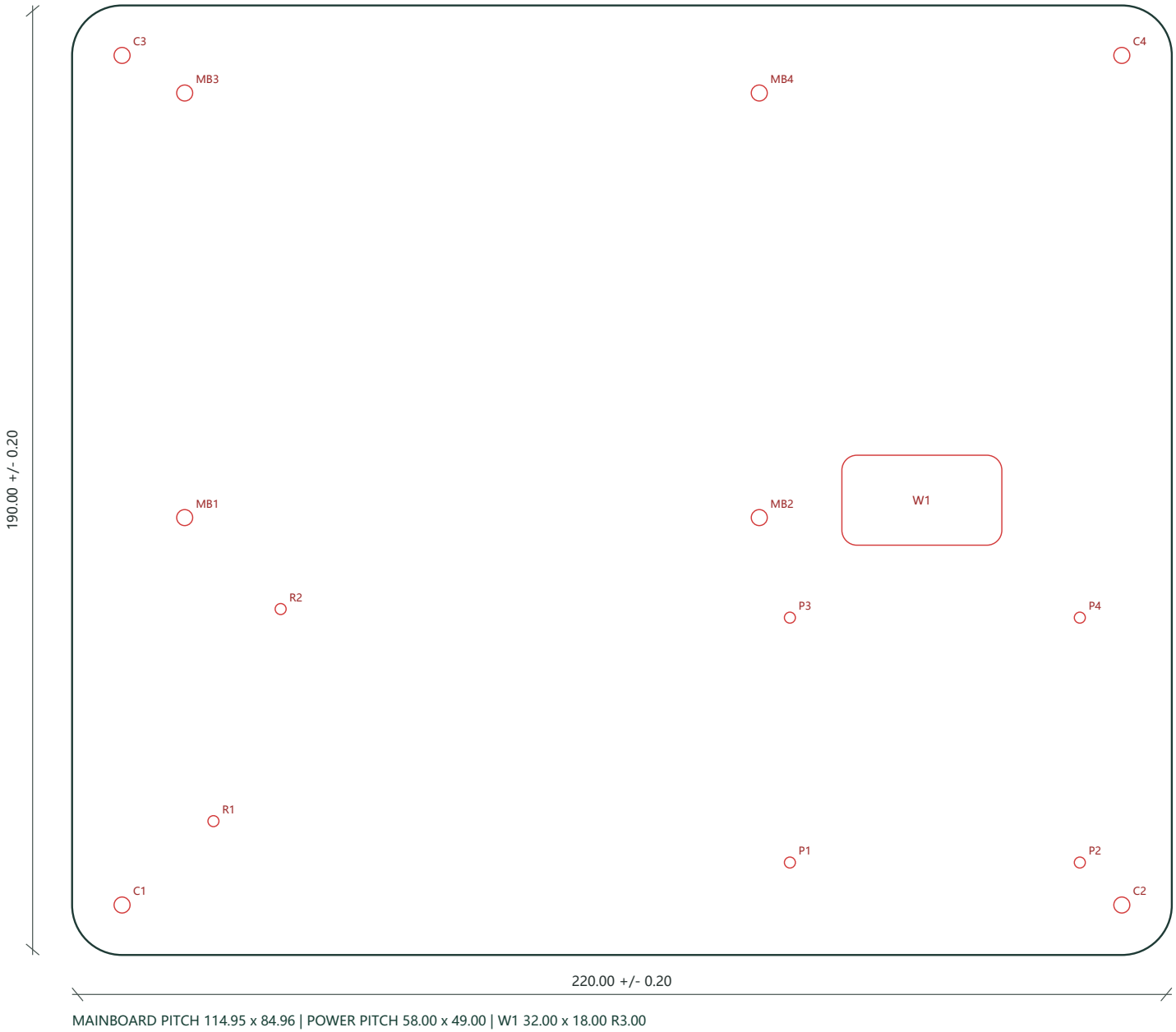
PAGE 1 - CUTTING / MACHINING DRAWING



MAINBOARD PITCH 114.95 x 84.96 | POWER PITCH 58.00 x 49.00 | W1 32.00 x 18.00 R3.00

MANUFACTURING DATA

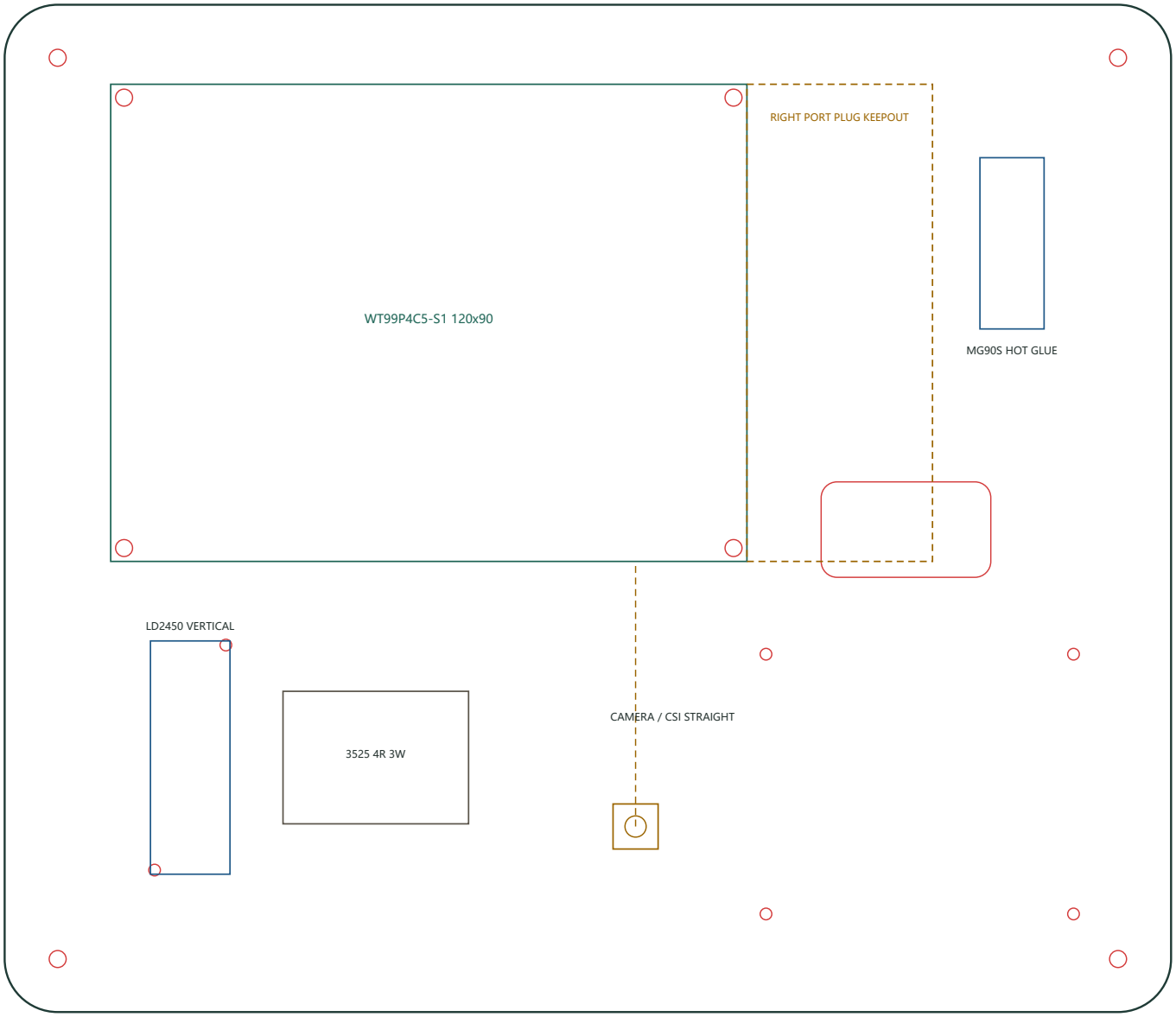
MATERIAL: CLEAR CAST PMMA
THICKNESS: 5.00 mm
CORNER: 4x R10.00
OUTLINE TOLERANCE: +/-0.20
HOLE/CUTOUT TOLERANCE: +/-0.10
PROCESS: CNC OR LASER
UNITS: mm SCALE: 1:1
DO NOT SCALE / FIT TO PAGE
DEBURR; KEEP PROTECTIVE FILM

ALL MACHINING COORDINATES ARE
DEFINED FROM FRONT VIEW.
DO NOT MIRROR CUTTING GEOMETRY.

FEATURE SCHEDULE

C1 X= 10.000 Y= 10.000 DIA 3.20
C2 X=210.000 Y= 10.000 DIA 3.20
C3 X= 10.000 Y=180.000 DIA 3.20
C4 X=210.000 Y=180.000 DIA 3.20
MB1 X= 22.525 Y= 87.520 DIA 3.20
MB2 X=137.475 Y= 87.520 DIA 3.20
MB3 X= 22.525 Y=172.480 DIA 3.20
MB4 X=137.475 Y=172.480 DIA 3.20
R1 X= 28.278 Y= 26.778 DIA 2.20
R2 X= 41.722 Y= 69.222 DIA 2.20
P1 X=143.600 Y= 18.500 DIA 2.20
P2 X=201.600 Y= 18.500 DIA 2.20
P3 X=143.600 Y= 67.500 DIA 2.20
P4 X=201.600 Y= 67.500 DIA 2.20
W1 C=(170.000,91.000) 32.00 x 18.00 R3.00 H

PAGE 2 - FRONT ASSEMBLY REFERENCE



FRONT ASSEMBLY

MAINBOARD: 4x M3, 8 mm STANDOFF

CSI CONNECTOR ON LOWER EDGE

LENS CENTER: X119.00 Y35.00

PCB EDGE TO LENS: APPROX. 50.00

CAMERA FLEX: STRAIGHT, NO SIDE TWIST

CAMERA: THIN DOUBLE-SIDED TAPE

RADAR: 15x44 VERTICAL, NOTCH UP

RADAR ANTENNA FACE: +Y / USER

RADAR: 2x M2 NOTCH CLAMPS

SPEAKER: 1 mm EVA TAPE

SERVO: HOT GLUE; SHAFT OUTWARD

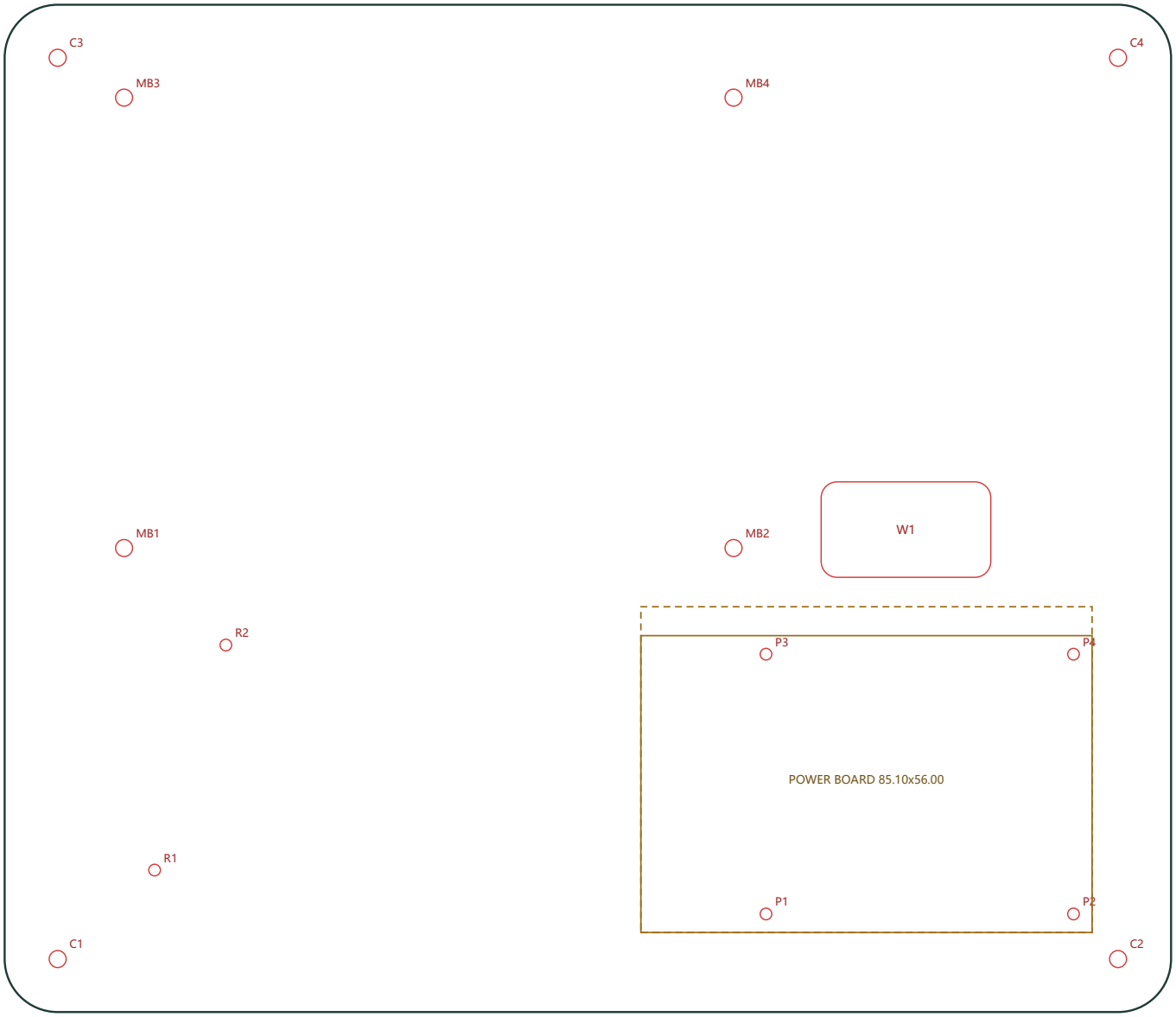
CAMERA/SPEAKER/SERVO: NO CUT HOLES

REFERENCE OUTLINES ARE NOT CUT GEOMETRY.

VERIFY PORT ACCESS BEFORE FINAL ASSEMBLY.

CAMERA FOCAL PLANE, MAINBOARD FRONT, AND RADAR ANTENNA PLANE FACE THE SAME DIRECTION (+Y).

PAGE 3 - REAR ASSEMBLY REFERENCE



REAR ASSEMBLY

POWER PCB ORIGIN: X120.00 Y15.00

PCB: 85.10 x 56.00 x 1.60

MAX PLAN ENVELOPE: 85.10 x 61.45

ASSEMBLED DEPTH: APPROX. 18.10

MOUNT: 4x M2, 10 mm STANDOFF

COMPONENT SIDE FACES REAR-OUT

HOLE PITCH: 58.00 x 49.00

W1 CLEARANCE ABOVE ENVELOPE: 5.55

ROUTE POWER/SIGNAL THROUGH W1

KEEP WIRES OUT OF CAMERA FLEX PATH

THIS PAGE SHOWS REAR COMPONENTS
USING FRONT-VIEW DATUM COORDINATES.
DO NOT FLIP X VALUES.
DO NOT MIRROR CUTTING GEOMETRY.

ALL MACHINING COORDINATES ARE DEFINED FROM FRONT VIEW. DO NOT MIRROR CUTTING GEOMETRY.

PAGE 4 - HOLE / CUTOUT COORDINATE TABLE

ID	TYPE	X	Y	WIDTH / DIA	HEIGHT / LENGTH	RADIUS	PURPOSE / REMARK
C1	CIRCULAR HOLE	10.000	10.000	DIA 3.20			M3 corner support
C2	CIRCULAR HOLE	210.000	10.000	DIA 3.20			M3 corner support
C3	CIRCULAR HOLE	10.000	180.000	DIA 3.20			M3 corner support
C4	CIRCULAR HOLE	210.000	180.000	DIA 3.20			M3 corner support
MB1	CIRCULAR HOLE	22.525	87.520	DIA 3.20			Mainboard M3
MB2	CIRCULAR HOLE	137.475	87.520	DIA 3.20			Mainboard M3
MB3	CIRCULAR HOLE	22.525	172.480	DIA 3.20			Mainboard M3
MB4	CIRCULAR HOLE	137.475	172.480	DIA 3.20			Mainboard M3
R1	CIRCULAR HOLE	28.278	26.778	DIA 2.20			LD2450 lower-left notch clamp; M2 standoff and nylon washer; washer overlap 0.5-0.8
R2	CIRCULAR HOLE	41.722	69.222	DIA 2.20			LD2450 upper-right notch clamp; M2 standoff and nylon washer; washer overlap 0.5-0.8
P1	CIRCULAR HOLE	143.600	18.500	DIA 2.20			Power board M2
P2	CIRCULAR HOLE	201.600	18.500	DIA 2.20			Power board M2
P3	CIRCULAR HOLE	143.600	67.500	DIA 2.20			Power board M2
P4	CIRCULAR HOLE	201.600	67.500	DIA 2.20			Power board M2
W1	ROUNDED RECTANGLE	170.000	91.000	32.00	18.00	R3.00	Cable and complete USB-A plug pass-through; HORIZONTAL; Deburr all edges

DIMENSION VERIFICATION AND CONFLICT CHECK

ITEM	OLD V2	SOURCE / USER INPUT	FINAL V3	RESULT
BACKPLATE	220x190x5 R10	User-confirmed	220x190x5 R10	MATCH
MAIN BOARD	90x120; old absolute position	WT99P4C5-S1 mechanical drawing	120x90 landscape at (20,85)	REPOSITIONED
CAMERA	4x 6x3.2 slots	Photo scale estimate authorized by user	Reference only; no cut	OLD SLOTS REMOVED
LD2450	4-hole bracket	Official 44x15 DXF/STEP; user clamp decision	15x44 vertical; 2 notch clamps	REBUILT
3525 SPEAKER	4-hole bracket	User-confirmed 35x25, 4 ohm, 3 W	Adhesive reference only	OLD HOLES REMOVED
MG90S	4-hole bracket plus door	MG90S drawing; user hot-glue decision	Hot glue; no cut	OLD HOLES REMOVED
POWER BOARD	58x49 pitch; wrong absolute location	85.10x56 drawing, 58x49, 3.50 edge offset	PCB at (120,15); 4x dia 2.20	REPOSITIONED
W1	dia 12 circle	User-confirmed full USB-A clearance	32x18 R3 at (170,91)	REBUILT

CAMERA DIMENSIONS ARE USER-AUTHORIZED PHOTO-SCALE ESTIMATES: +/-1 mm LATERAL, +/-2 mm LENGTH/LENS POSITION.

CAMERA GEOMETRY IS ASSEMBLY REFERENCE ONLY AND DOES NOT CONTROL ANY CUT FEATURE.

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